

$$1) \Delta E = \left(\frac{1}{2} m v^2 \right) \cdot \sqrt{\left(\frac{\Delta m}{m} \right)^2 + \left(\frac{2 \Delta v}{v} \right)^2}$$

$$\Delta g = \left(\frac{GM}{r^2} \right) \cdot \sqrt{\left(\frac{\Delta G}{G} \right)^2 + \left(\frac{\Delta M}{M} \right)^2 + 2 \left(\frac{\Delta r}{r} \right)^2}$$

$$\Delta v = \left(\frac{4 \pi r^3}{3} \right) \cdot \sqrt{\left(\frac{3 \Delta r}{r} \right)^2}$$

$$\Delta k = \left(\frac{\sqrt{2} m v}{h} \right) \cdot \sqrt{\left(\frac{\Delta m}{m} \right)^2 + \left(\frac{\Delta v}{v} \right)^2 + \left(\frac{\Delta h}{h} \right)^2}$$

$$* \Delta f = (c^x) \cdot \sqrt{\left(x \cdot \frac{\Delta c}{c} \right)^2}$$

$$2a) x = \frac{1}{2} a t^2$$

$$b) \Delta a = 2(\Delta a) \approx 0,004 \text{ m/s}^2$$

$$a = 2A(1,318 \pm 0,004) \text{ m/s}^2$$

$$A = \frac{1}{2} \cdot a$$

$$2A = a$$

Gráficos na próxima
página

$$a_0 = 0,0143 \pm 0,02$$

$$a_1 = -0,0199 \pm 0,04$$

$$a_2 = 0,6664 \pm 0,01$$

$$f: a_0 + a_1 x + a_2 x^2$$

$$\Delta a = 2(\Delta a_2 + \Delta a_1 + \Delta a_0) \approx 0,014$$

$$a = \frac{2y}{x^2} = (1,3 \pm 0,2) \text{ m/s}^2$$

$$3a) A(t) = A_0 e^{-\frac{bt}{2m}} \quad f(x) = be^{ax} \Rightarrow y = Ax + B$$

$$\ln A(t) = \ln A_0 \cdot \ln e^{-\frac{bt}{2m}}$$

$$\ln A(t) = \ln A_0 \cdot \left(\frac{-bt}{2m} \right)$$

$$\ln A(t) = \ln A_0 - \left(\frac{bt}{2m} \right)$$

$$\begin{cases} y = \ln A(t) \\ B = \ln A_0 \\ A = -b/2m \\ x = t \end{cases}$$

$$b) A = (-0,0029 \pm 0,0003)$$

$$B = (0,015 \pm 0,001)$$

$$y = 0,029 \cdot x + 0,015$$

$$A(t) = 0,015 \cdot \left(\frac{0,0029 t}{2} \right)$$

2a)

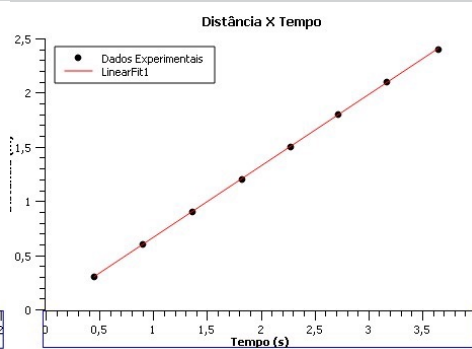
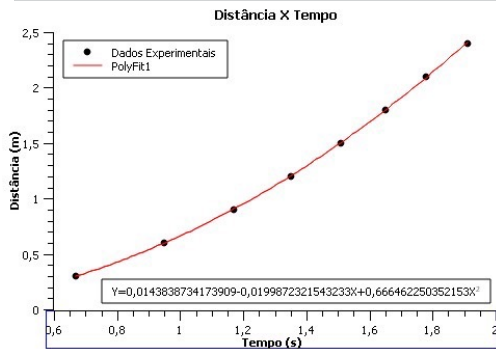
[sábado, 13 de abril de 2024 20:57:56 E. South America Standard Time
Polynomial fit of dataset: Table1_1, using function: $a_0+a_1*x+a_2*x^2$
Y standard errors: Unknown
From x = 0,67 to x = 1,91
 $a_0 = 0,0143838734173909 \pm 0,0252034000596034$
 $a_1 = -0,0199872321543233 \pm 0,0410505398024352$
 $a_2 = 0,666462250352153 \pm 0,0156443409653271$
Chi^2 = 0,000204966501713505
R^2 = 0,99998836247183

Plot: "Graph1"

b)

[sábado, 13 de abril de 2024 21:51:52 E. South America Standard Time
Linear Regression fit of dataset: Table1_3, using function: $A*x+B$
Y standard errors: Unknown
From x = 0,4489 to x = 3,6481
 B (y-intercept) = $0,00235464933793983 \pm 0,004654555246138$
 A (slope) = $0,658918756702857 \pm 0,00202732373832159$
Chi^2 = 0,00021468458134835
R^2 = 0,999988306940014

Plot: "Graph4"



3b)

[sábado, 13 de abril de 2024 21:13:55 E. South America Standard Time
Linear Regression fit of dataset: Table2_1, using function: $A*x+B$
Y standard errors: Unknown
From x = 0,5 to x = 5
 B (y-intercept) = $0,01454 \pm 0,000989070576885495$
 A (slope) = $-0,00293818181818182 \pm 0,000318806313558465$
Chi^2 = $1,67701818181818e-05$
R^2 = 0,972604009183877

Plot: "Graph2"

